Fiscal Year:	FY 2010	**Task Last Updated:**	FY 03/19/2010
PI Name:	Yu, Yongjia Ph.D.		
Project Title:	Impact of HZE Particles Exposure on Proliferation and Differentiation of Human Neural System Cells		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77555-0884	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Radiation Biology and Shielding 03-OBPR-07
Start Date:	05/15/2004	**End Date:**	11/30/2009
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 11/30/2009 per J. Dardano/JSC (5/09) NOTE: Received NCE to 5/31/2009 per J. Dardano/JSC (5/08) End date changed back to 6/1/2008 per PI (5/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Chuang, Eric (National Institutes of Health) Morrill, Steve (UTMB) Wu, Ping (UTMB)		
Grant/Contract No.:	NNJ04HD81G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Space radiations such as HZE particles pose a great risk of damaging the central nerve system (CNS) of astronauts. Human neural stem cells (hNSCs) are of particular interest since functional NSCs in adult brain are potential targets of HZE damage, and because exogenous hNSCs may provide potential therapeutic source for replacing lost neurons in radiation-damaged brain. The long-term objective of this project is to study the mechanisms underlying HZE particles-induced adverse impact on the proliferation and differentiation of a primary human fetal neural stem cell line, K048. K048 retains self-renewal and multipotential differentiation capabilities in vitro and in vivo. Moreover, a novel in vitro priming procedure appears to direct K048 cells to adopt neuronal differentiation fate. We will test two hypotheses:(1) HZE particles disrupt proliferation and differentiation of human NSCs by activating apoptotic and genotoxic-response pathways. (2) Inhibitory neurogenic micro-environmental cues produced by HZE in rat brain can be overcome by this novel in vitro priming prior to transplantation into brain. We will pursue three specific aims: (1) Determine the impact of 56Fe and gamma-ray irradiation on proliferation and differentiation of K048 cells in vitro. (2) Determine gene expression patterns and apoptotic responses of proliferating and differentiating K048 cells following 56Fe and gamma-ray irradiation. (3) Determine whether in vitro primed K048 cells can differentiate into neurons when grafted into neurogenic and non-neurogenic regions of rat brains damaged by 56Fe and gamma-rays. The results of this project will contribute to a better understanding of cellular and molecular mechanisms of damage to the CNS following irradiation by HZE particles.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In this project, we propose to examine the impact of HZE and low LET radiations on the proliferation and differentiation of K048 human neural stem cell line. The K048 cell model is a unique experimental system not only because of its human origin, but also because this line is not transformed and retains the characteristics of normal neural stem cells. Using this system we will address directly to one of the high priority topics listed in NRA 03-OBPR-07 Research Announcement, namely “What are the cellular and molecular mechanisms of damage to the CNS following irradiation by HZE particles?”. The proposed research is also within the scope of “Critical Path Roadmap”, regarding risk number (39) “Degenerative tissue diseases including damage to the central nervous system, cataracts, circulatory diseases, or other late non-cancer risks”, and risk number (41) “Early or acute effects from radiation exposure”. Specifically, our proposed studies address Critical Question (10.05) in radiation, “Are there unique biological effects associated with HZEs?”
Task Progress:	Most of the experiments we put in the original proposal have been completed. These experiments have led to the following key findings: 1. Within a dose range of 0.1 to 2 Gy, 56Fe irradiation inhibits neuronal differentiation of K048 stem cells in a dose-dependent manner. Gamma-radiation has a similar effect. 2. Inhibition of neuronal differentiation of K048 cells by 56Fe-irradiation is persistent. 3. 56Fe-irradiated K048 stem cells generate more astroglial cells during differentiation, while gamma-rays have little effect on astroglial differentiation of K048. 4. Effect of 56Fe irradiation on astroglial differentiation diminishes gradually over time. 5. K048 neural stem cells respond to 56Fe-irradiation by predominantly down regulating expression of many genes involved in multiple signal transduction pathways. 6. Cranial irradiation with 56Fe induced microglia activation that lasted for at least 10 weeks in rat. This microenvironment inhibits neuronal differentiation of grafted non-irradiated K048 cells. However, ex vivo pretreatment of these cells with bFGF, heparin and laminin (FHL) appears to be able to counter inhibitory cue and enhance their neurogenesis in vivo.
Bibliography Type:	Description: (Last Updated: 03/19/2010)
Abstracts for Journals and Proceedings	Yu Y, Cittelly D, Wu P, Cornforth MN. "Inhibition of Human Neural Stem Cell Differentiation by gamma-rays." Presented at Bioastronautics Investigators' Workshop, Galveston, TX, January 10–12, 2005. Abstracts, Bioastronautics Investigators' Workshop, Galveston, TX, January 10–12, 2005. Online Pub: http://www.dsls.usra.edu/meetings/bio2005/pdf/SRPosters/2075Yu.pdf , Jan-2005
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Abstracts for Journals and Proceedings	Yu Y, Gao JL, Gao Y, Dunn T, Wu P. "Neuronal differentiation of grafted human neural stem cells in 56Fe-irradiated rat brain." Presented at NASA Human Research Program investigators' workshop, League City, Texas, February 4-6, 2008 Abstracts, NASA Human Research Program Investigators' Workshop, League City, Texas, February 4-6, 2008. , Feb-2008
Abstracts for Journals and Proceedings	Yu Y, Wu P. "Differential regulation of Neuregulin-1 expression in human neural stem cells by 56Fe ions and gamma-rays." Presented at the 19th Annual NASA Space Radiation Investigators' Workshop, Philadelphia, PA, June 30-July 2, 2008 Abstract. 19th Annual NASA Space Radiation Investigators' Workshop, Philadelphia, PA, June 30-July 2, 2008 , Jun-2008
Abstracts for Journals and Proceedings	Yu Y, Dunn T, Wu P. "Interaction between neural stem cells and astroglial cells following irradiation." Presented at NASA Human Research Program investigators' workshop, League City, Texas, February 2-4, 2009. Abstract. NASA Human Research Program investigator's workshop, League City, Texas, February 2-4, 2009. , Feb-2009